BLOOD SMEAR DIAGNOSTIC SUMMARY

Patient: Anonymous	ID: N/A	Date: 2025-12-22 19:52

■ PARASITIC INFECTIONS

Test	Result	Finding	Clinical Interpretation
Malaria (P. falciparum/vivax)	✓ NOT DETECTED	0 structures detected	No parasite-like structures identified in analyzed regions.
Method	YOLO Object Detection	Stages: None	Negative screening. Clinical suspicion requires confirmation with microscopy.

■ HEMOGLOBINOPATHIES (Sickle Cell & Thalassemia)

Condition	Result	Findings	Interpretation
Sickle Cell	✓ NEGATIVE	0 cells (0.0%)	No sickle cells detected.
Thalassemia	✓ NEGATIVE	0 indicators (10.0%)	No significant thalassemia indicators.
Next Steps			Sickle: Normal finding Thalassemia: Normal RBC morphology

■ ANEMIA SCREENING (RBC Morphology)

Anemia Type	Indicator	Findings	Interpretation
Iron Deficiency (Microcytic)	✓ NORMAL SIZE	0.0% microcytes	Normal RBC size distribution.
Size Variation (Anisocytosis)	✓ NORMAL	Size CV: 0.0%	Normal size variation.

■ INFECTION INDICATORS (WBC Differential Count)

Cell Type	Count	%	Normal Range	Clinical Significance
Neutrophil	5	50.0%	40-70%	✓
Lymphocyte	3	30.0%	20-40%	✓
Monocyte	2	20.0%	2-10%	Chronic infection, TB, malaria
Eosinophil	0	0.0%	1-5%	✓
Basophil	0	0.0%	0-2%	✓

Clinical Interpretation: ↑ Monocyte (20.0%): Chronic infection, TB, malaria

■■ LEUKEMIA / MALIGNANCY SCREENING

Type	Result	Findings	Clinical Action
Blast Cells (N/C Ratio)	■■ DETECTED	2 blast-like cells (20.0%) Avg N/C: 0.65	URGENT: >20% blast-like cells detected. Immediate hematology referral for bone marrow bio
ALL (Acute Lympho.)	✓ NEGATIVE	Lymphocytes: 30.0%	No lymphocytic predominance.
CML (Chronic Myeloid)	■■ SUSPICIOUS	Neutrophils: 50.0%, Monocytes: 20.0%	Myeloid predominance may indicate CML (Chronic Myeloid Leukemia). Hematology referral re

✓ **N/C Ratio Analysis Enabled:** Nucleus-to-cytoplasm ratio calculated from segmentation masks. Blast cells (N/C >0.80) flagged for manual review. Atypical morphology still requires hematologist confirmation.

Methods: YOLO11n object detection (malaria), ResNet34 classification (WBC, RBC), Instance segmentation (RBC morphology). Generated: 2025-12-22 19:52:39